Roll No.....

End Semester Examination 2025

Name of the Program: M.TECH.(CSE)
Name of the Course: Mobile Computing

Semester: 1ST
Course Code: MCS 102

Time: 03 Hours
Note:

Maximum Marks: 100

(i) Answer **all the questions** by choosing ***any two of the sub questions***.
(ii) Each question carries 20 marks

Q1	(20 marks)	CO1
a)	What is meant by handoff interval and handoff region? Explain their usefulness with appropriate diagram.	
b)	Explain HLR and VLR concept for location management in mobile communication. What is role of CoA? How is it assigned?	
c)	Explain the task of MAC layer? Name some protocols with brief description for the purpose of service provision at this layer.	
Q2	(20 marks)	CO2
a)	What do you mean by multihopping? Explain multi cluster architecture and problems related to this architecture.	
b)	Discuss Design architecture of Coda system. How the interprocess-communication took place in this file system.	
c)	Mention the two basic important ideas on which Rover toolkit works. Explain the working mechanism of this toolkit.	
Q3	(20 marks)	CO3
a)	What is pointer forwarding strategy? What are required schemes, which are used to reduce signalling traffic?	
b)	Show clearly how IP-in-IP tunneling is used in mobile IP.	
c)	What is the need for energy efficient indexing and how can we manage broadcast data on air using indexing method?	
Q4	(20 marks)	CO4
a)	Discuss about the Mobile agent. Describe the model for mobile agent based computing. Also discuss the possible ways for providing security to mobile agents.	
b)	How mobile agents are helpful in the parallel computing and data collection.	
c)	Discuss about the Cyber bulling. In order to cyber security Intrusion Detection became an important aspect. Therefore, Describe the key methods for Intrusion Detection.	
Q5	(20 marks)	CO5
a)	Classify the Adhoc network? What are the main functions and the design issues of medium access control layer in Mobile Adhoc Networks?	
b)	Is DSR protocol is suitable for high mobility? Explain some disadvantage DSR routing protocol.	
c)	Indicate the purpose of 'Echo' and 'Hello' packet in the routing. Compare and contrast the routing mechanisms in mobile IP and AODV.	